INCH-POUND

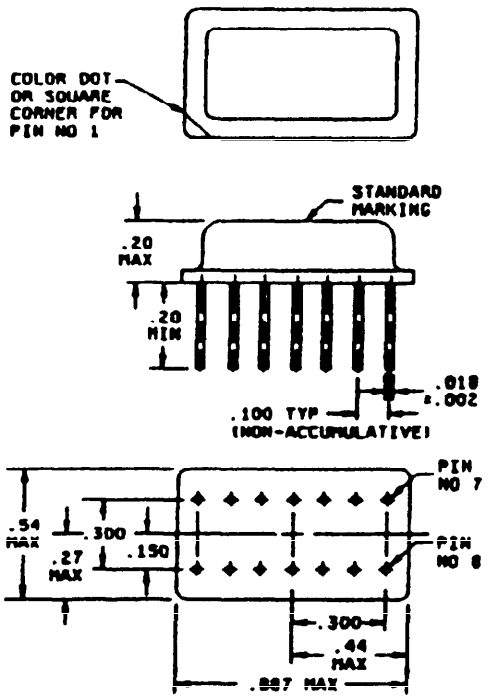
MIL-O-55310/14D
15 March 1994
SUPERSEDING
MIL-O-55310/14C
16 February 1984

MILITARY SPECIFICATION SHEET

OSCILLATOR, CRYSTAL CONTROLLED, TYPE 1 (CRYSTAL OSCILLATOR (XO)),
0.1 Hz THROUGH 25 MHz, HERMETIC SEAL, SQUARE WAVE, TTL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-O-55310



Pin number	Function
1	NC
2	NC
3	NC
4	B+
5	OUTPUT
6	NC
7	B- (GND/CASE)
8	NC
9	NC
10	NC
11	NC
12	NC
13	NC
14	NC

Inches	mm
.002	0.05
.018	0.46
.100	2.54
.150	3.81
.20	5.1
.27	6.9
.300	7.62
.44	11.2
.54	13.7
.887	22.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. All pins with NC function may be connected internally and are not to be used as external tie points or connections.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction: See figure 1.

Mounting: See figure 1.

Terminals: See figure 1.

Seal: Hermetic in accordance with MIL-O-55310, maximum leakage rate 5 x 10^{-8} atm cc/s.

Weight: 0.5 ounce maximum.

Construction technology: Hybrid construction.

Calibration: Manufacturer calibrated.

Screening: In accordance with MIL-O-55310, class B or S, as applicable.

Temperature:

Operating: See table I.

Storage: -62°C to +125°C.

Oscillator load: 1 to 10 standard TTL loads. A TTL unit load is defined as: 1.6 mA sink, 0.04 mA source, and 2 pF capacitance.

Output waveform: Symmetrical square wave.

Supply voltage: See table I.

Input current: At designated supply voltage (see table I).

Output frequency: Frequency as designated at time of acquisition (see table I).

Output voltage: At designated TTL load (see table I).

Logic 1: See table I.

Logic 0: See table I.

Rise and fall times: See table I.

Duty cycle: See table I.

Initial accuracy at reference temperature (up to 30 days after shipment): See table I.

Initial frequency-temperature accuracy (one-half temperature cycle): Quality assurance provisions applicable. 1/

Frequency-temperature tolerance (one-half temperature cycle, referenced to frequency measured at +23°C, immediately prior to starting of the test): See table I. Measurements taken at 10 equispaced increments over the specified operating temperature range. 1/

1/ For the purpose of transitioning this device to MIL-O-55310B, 'Frequency stability versus temperature' has been renamed 'Frequency-temperature tolerance'. The quality assurance requirements of 'Initial frequency-temperature accuracy (one-half temperature cycle)' shall apply except that frequency measurements shall be referenced to the frequency measured at +23°C (f_{ref}) instead of to the nominal frequency (f_{nom}).

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TABLE I. Dash numbers and operating characteristics.

Dash number	Output frequency range	Supply voltage	Input current (max) 2/	Output voltage		Rise and fall times (max)	Initial accuracy ppm at +23°C	Duty cycle at 1.4 V	Frequency-temperature tolerance (ppm) 1/		
				Logic: 1 (min)	Logic: 0 (max)				-55°C to +125°C	-55°C to +105°C	-20°C to +70°C
									A	B	C
		V dc	mA			ns		percent			
01	0.1 Hz to 1 kHz	+5 ±0.5	158	2.4 V dc at 400 µA source	0.5 V dc at 16 mA sink	15	±15	45 to 55	±50	±40	±30
02	1 kHz to 150 kHz	"	158	"	"	"	"	"	"	"	"
03	150 kHz to 300 kHz	"	94	"	"	"	"	"	"	"	"
04	300 kHz to 600 kHz	"	94	"	"	"	"	"	"	"	"
05	600 kHz to 2.5 MHz	"	50	"	"	"	"	"	"	"	"
06	2.5 MHz to 5 MHz	"	40	"	"	"	"	"	"	"	"
07	5 MHz to 10 MHz	"	30	"	"	"	"	"	"	"	"
08	10 MHz to 15 MHz	"	20	"	"	"	"	40 to 60	"	"	"
09	15 MHz to 25 MHz	"	20	"	"	5	"	40 to 60	"	"	"

1/ Temperature range A applicable for class B oscillators only.

2/ Maximum input current for no load condition. Actual configuration of TTL loads must be added to determine power supply requirements.

Frequency-voltage tolerance: ±2 ppm maximum for a ±10 percent change in supply voltage. Measurements taken at reference temperature and operating temperature range end points.

Frequency aging: Measurements shall be taken at +70°C ±0.2°C at intervals of not more than every 72 hours for 30 days minimum.

±5 ppm per year maximum

±0.7 ppm per 30 days
±1.5 ppm per 90 days

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Terminal strength: MIL-STD-202, method 211, test condition C.

Applied force: 2 pounds each terminal for 10 seconds.

Bends: Five at 45° each.

Frequency-environmental tolerance: Not applicable.

Vibration, sinusoidal: In accordance with MIL-O-55310 and MIL-STD-202, method 204.

Nonoperating: Test condition D.

Operating: Test condition D.

Ambient pressure:

Nonoperating: In accordance with MIL-O-55310.

Operating: MIL-STD-202, method 105, test condition C.

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash and coded alphas, and numeric number. See example:

EXAMPLE

M55310/14- S 01 A XXXXXXXX

M prefix and specification sheet number

Product assurance level (S or B)

Dash number (see table I)

Operating temperature range (A, B, or C) (see table I)

Frequency

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0677-10)

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